

ML Analysis Plan for Ocean Carbon Sink Study

1. Data Sources to Gather

Based on what the paper itself used, the real-world observational data you'd want to pull together includes:

- **Keeling Curve data** (atmospheric pCO₂ over time, NOAA/Scripps)
 - **Sea Surface Temperature** (COBE-SST or OISST datasets)
 - **Ocean salinity** (World Ocean Atlas or Argo float data)
 - **Sea-air CO₂ flux** (SOCAT or Landschützer's gridded pCO₂ dataset, already cited in the paper)
 - **Wind speed data** (NCEI blended wind products, also cited in the paper)
 - **Amazon River discharge** records (ANA Brazil / HYBAM observatory)
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2. ML Tasks & Recommended Algorithms

A. Prediction — Future CO₂ and Salinity Trends The goal here is to learn the temporal dynamics the box model captured with ODEs, but purely from data.

- **LSTM / GRU (Long Short-Term Memory networks)** — ideal since CO₂ and salinity are time series with long-range dependencies
- **Temporal Fusion Transformers** — handles multiple ocean variables simultaneously
- **Facebook Prophet** — simpler baseline for trend + seasonality decomposition
- Target variables would be C_i (aqueous CO₂) and S_i (salinity) per ocean basin

B. Validation — Do ML predictions agree with the box model? Here ML acts as an independent check on the paper's ODE results.

- Train ML models on observational data from 2000–2015, predict 2015–2020
- Compare ML-predicted CO₂/salinity trends against what the box model simulated
- Agreement would strongly back the paper; disagreement highlights where the simplifying assumptions of the box model break down
- **XGBoost / Random Forest regressors** work well here as interpretable baselines alongside the deep learning models

C. Pattern Discovery — What the Box Model Missed
The paper openly acknowledges limitations: no Arctic

Ocean, no terrestrial runoff, no seasonal depth variation.
ML can probe these gaps.

- **Clustering (K-Means, DBSCAN, or Gaussian Mixture Models)** applied to gridded pCO₂ + SST + salinity data to find natural ocean sink/source groupings — compare these clusters against the paper's 5-box structure to see if the boundaries hold up
- **Anomaly detection (Isolation Forest, Autoencoders)** to find unusual CO₂ flux events the box model wouldn't capture, like El Niño-driven Pacific variability
- **PCA / UMAP** for dimensionality reduction across all ocean variables to visualize which features co-vary most strongly

3. Specific Hypotheses ML Can Test

These map directly to the paper's claims:

Paper Claim	ML Test
Southern Ocean is the strongest CO ₂ sink	Clustering should isolate SO as a distinct high-absorption region
North Atlantic becomes most saline over time	LSTM trained on salinity should predict NA diverging upward

Amazon freshwater flux is a key control parameter

SHAP values on regression model should rank Amazon discharge high

Indian Ocean salinity is near a bifurcation point

Anomaly detection near IO salinity fixed point